



Technical University of Moldova
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INTERNSHIP REPORT

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*By signing the cover page of this report, each student confirms their active involvement in both the development of the project and the writing of the report. Students who did not contribute meaningfully are not permitted to sign.

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ABSTRACT

You probably have to follow the structure of this first paragraph.

The thesis titled **Fission in the Frontend: Building a Nuclear Reactor in JavaScript** was developed by student LastName FirstName as a bachelor's project at the Technical University of Moldova. It is written in English and contains an introduction, 4 chapters (Domain Analysis, Requirements Specification, System Design, System Implementation), a list of figures, a list of tables, conclusion, bibliography, and appendices.

Keywords: JavaScript, nuclear reactor, frontend engineering, asynchronous meltdown, satire

In this groundbreaking study, we explore the totally safe and not-at-all terrifying proposition of building and operating a fully functional nuclear reactor system entirely in JavaScript. Historically relegated to the humble browser and the occasional web app that crashes if you press Backspace too hard, JavaScript is the ideal language for controlling a high-risk, high-complexity critical infrastructure system like a nuclear power plant. Leveraging cutting-edge technologies like `setTimeout()`, `localStorage`, and a loosely defined global state, we demonstrate how a single-page web app can reach critical mass—both figuratively and literally. With asynchronous core cooling (using Promises), CSS-based hazard lights, and a Node.js-powered Geiger counter API (because why not), our system boasts at least 60% uptime and 40% existential dread. This paper is a satirical ode to the modern web stack and its uncanny ability to appear everywhere it absolutely should not.

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INTRODUCTION

Changes in technology continue to alter possibilities for learning and create new challenges for pedagogy. Over the last two decades, colleges and universities adapted and responded to the Internet, email, chat, and instant messaging, course management software, podcasts, personal digital assistants (PDAs), and much more. The growing use of mobile technology at colleges and universities is the most current trend forcing educators to evaluate the merits and limitations of new technology. The Internet is particularly among students 12–25-years. Further, web technology figures prominently in the future of higher education, particularly in its integration into teaching and learning. This project is an web application used to help both teachers, students for communication, searching and exchanging information on any topic, from hobbies to news, offering users the opportunity to join thematic channels to discuss content and find solutions to their problems.

1 DOMAIN ANALYSIS

The absence of a platform for students to communicate and share knowledge, coupled with the limitations of the end-of-semester SIMU feedback system, restricts timely academic improvements and leaves prospective students with insufficient authentic information about university life.

1.1 Problem Definition

In Moldova, there is a significant gap in platforms that enable students to communicate, share knowledge, and exchange experiences, unlike in other countries where such platforms are common. The primary mechanism for student feedback, the end-of-semester evaluation platform (SIMU), is limited to collecting feedback once per semester without facilitating interaction or discussion among students. This restricts timely improvements to the academic experience. Additionally, students perceive a lack of anonymity in SIMU, fearing academic reprisal, which discourages honest and constructive feedback. Furthermore, prospective students rely on curated university marketing materials and anecdotal stories, which fail to provide authentic insights into daily student life, leaving critical questions about courses and campus experiences unanswered. A solution is needed to create a platform that fosters continuous, anonymous, and transparent communication and knowledge-sharing among students, enhancing the academic environment and providing reliable information for prospective students.

- **Infrequent Feedback Collection:** The SIMU platform collects feedback only once per semester, at its conclusion. This timing prevents instructors and administrators from implementing timely changes to address student concerns, rendering the feedback ineffective for improving the ongoing academic experience.
- **Perceived Lack of Anonymity:** Despite assurances of anonymity, many students fear that their critical feedback could be traced back to them. This apprehension, stemming from concerns about potential academic reprisal from professors, discourages honest and constructive criticism, undermining the quality and candor of the feedback provided.
- **Limited Information for Prospective Students:** Their primary sources are official marketing materials and anecdotal stories. While helpful, marketing content is designed to present the university in the best possible light and often fails to capture the day-to-day realities of student life. This leaves critical questions unanswered about course difficulty, professor teaching styles, campus culture, and the true student routine.

To address these challenges, our proposed solution, the Student Forum application, aims to provide a platform that facilitates continuous, anonymous, and transparent feedback from students. This platform will

empower students to share honest insights and experiences in real time, fostering a more responsive academic environment and providing prospective students with a clearer, more authentic picture of university life.

1.2 Some Technical Concept

The next section can be an introduction to some technical concepts that your topic requires. Use diagrams, either your own or from the internet (in which case you have to cite them).

Floats (figures, tables and code listings) are not allowed to be at the top of the page, there should be at least a few lines separating them from the top. Also, they may not touch each other, there should be at least a few lines separating them. Look up how to use `[!hb]` after `\begin{figure}`, this way Latex will aid you in placing them according to these rules.

1.2.1 Subsection Example

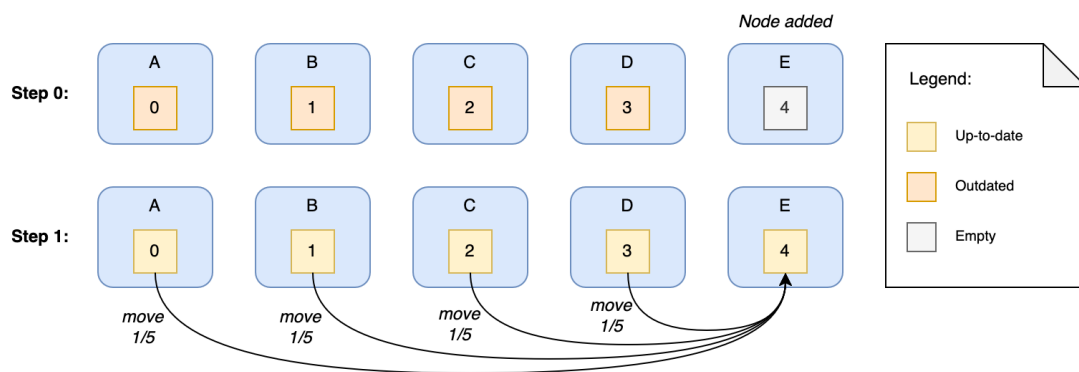


Figure 1.1 - The Diagram Captions Are Also Title Case [greenwade93]

A figure should appear in the same section it's referenced in, so be careful that Latex doesn't place it in the next section because it couldn't fit in the current one. It can be placed before it is referenced though (I think, correct this if I'm wrong). This is one reason why sections should span at least one page, so that you have enough room to place the floating figures within it. If your sections are too small, simply merge them.

Notice how figure ?? couldn't fit on the page right after the section title where it was defined, so it was moved to the next page instead. But now it's at the top of the page, which UTM doesn't allow. So you have to move the definition of the figure somewhere else, in this case after this paragraph. Setting `[!hb]` on the figure would make it go to the bottom of the page, leaving a big gap between this text and the figure, so that's not a solution.

1.3 How to write Mathematics

L^AT_EX is great at typesetting mathematics. Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed random variables with $E[X_i] = \mu$ and $\text{Var}[X_i] = \sigma^2 < \infty$, and let Equation ??:

$$S_n = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_i^n X_i \quad (1.1)$$

denote their mean. Then as n approaches infinity, the random variables $\sqrt{n}(S_n - \mu)$ converge in distribution to a normal $\mathcal{N}(0, \sigma^2)$.

1.4 How to add Citations and a References List

You can simply upload a .bib file containing your BibTeX entries, created with a tool such as JabRef. You can then cite entries from it, like this: [greenwade93]. Just remember to specify a bibliography style, as well as the filename of the .bib. You can find a video tutorial here to learn more about BibTeX.

If you have an upgraded account, you can also import your Mendeley or Zotero library directly as a .bib file, via the upload menu in the file-tree.

1.5 Existing Solutions

Here is an example of how to format a Table ??, you might need it to compare existing solutions. Formatting tables is probably the most difficult thing in LaTeX.

Table 1.1 - Example Table That Fits Page Width

Column 1	Column 2	Column 3
Long wrapped content that fits nicely in the cell and breaks into multiple lines	Another wrapped text entry to demonstrate alignment	Final column with enough text to require wrapping too
Short entry	Medium text content here	More wrapped content

2 SYSTEM DESIGN

Here is a good place to put your UML diagrams like Component, Deployment, State, Sequence. Basically everything that explains how your *particular* system works.

3 SYSTEM IMPLEMENTATION

Paragraph before code.

Listing 3.1 - Main Reactor Loop [Sarkar2020OAuth]

Paragraph after code (check the spacing).



Figure 3.1 - One more frog for good measure

CONCLUSIONS

APPENDIX A

Some code